BUILD A GREENER WORLD
CREATE A BETTER LIFE

Linyang Power Atlantic 4MWh BESS

Degradation Curve

PREPARED BY:

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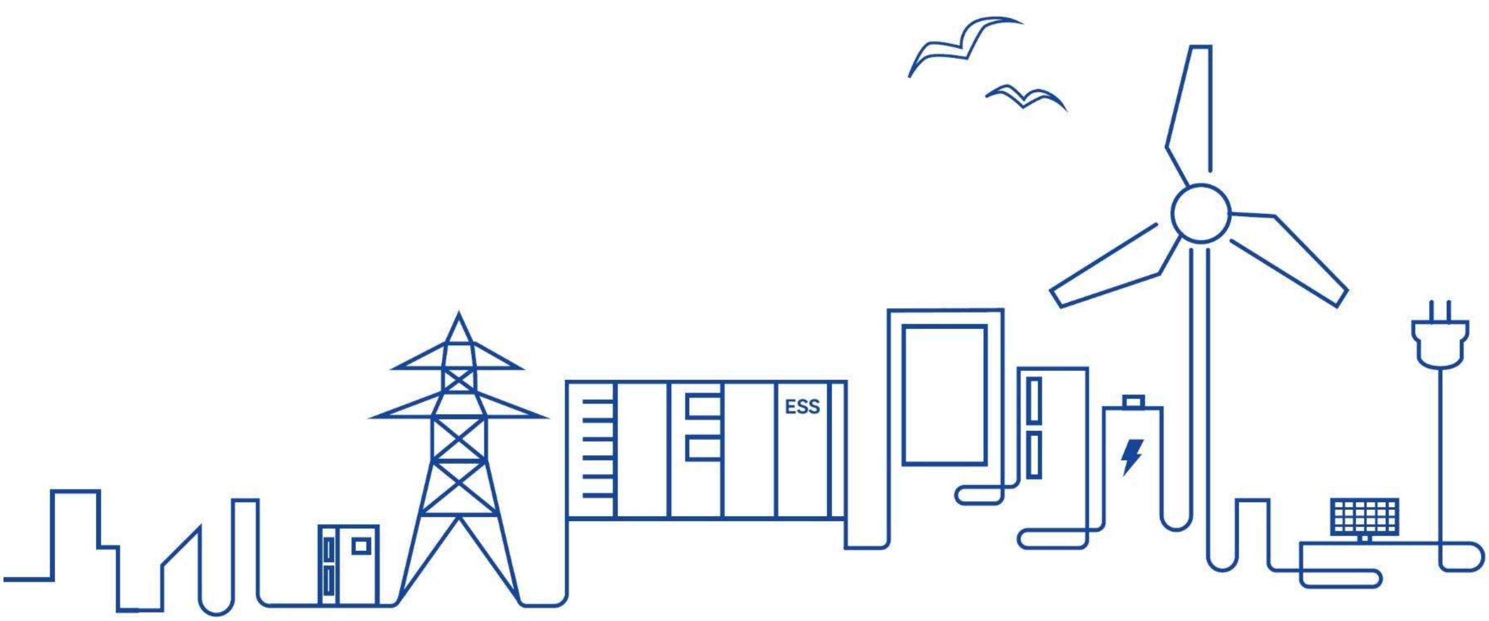

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● Abbreviations

BESS	Battery Energy Storage System
COD	Commercial Operation Date
CPD	Cycle per Day
EOL	End of Life
SOH	State of Health

1. BESS Degradation Curve Description

1.1 Description

The degradation curve of a BESS describes the variation of the energy storage capacity of an energy storage system over time. As time passes, the energy storage capacity of the BESS degrades until it reaches the EOL set by the system.

For the purposes of this paper, we consider SOH at COD time to be 98.5%. COD time is considered as 2 + 6 months after production, i.e. 2 months for shipment and max 6 months for installation and commissioning.

2 BESS Degradation Curve

2.1 Operating Condition: 0.25P, 1 Cycle per Day

Life Cycle Assessment of BESS (1 Cycle/day) - 0.25P

Year	SOH	Cycles
0	98.50%	0
1	95.39%	365
2	92.89%	730
3	90.82%	1095
4	89.36%	1460
5	88.12%	1825
6	86.72%	2190
7	85.56%	2555
8	84.38%	2920
9	83.16%	3285
10	82.22%	3650
11	81.16%	4015
12	80.10%	4380
13	79.04%	4745
14	77.98%	5110
15	76.91%	5475
16	75.85%	5840
17	74.79%	6205
18	73.73%	6570
19	72.67%	6935
20	71.60%	7300

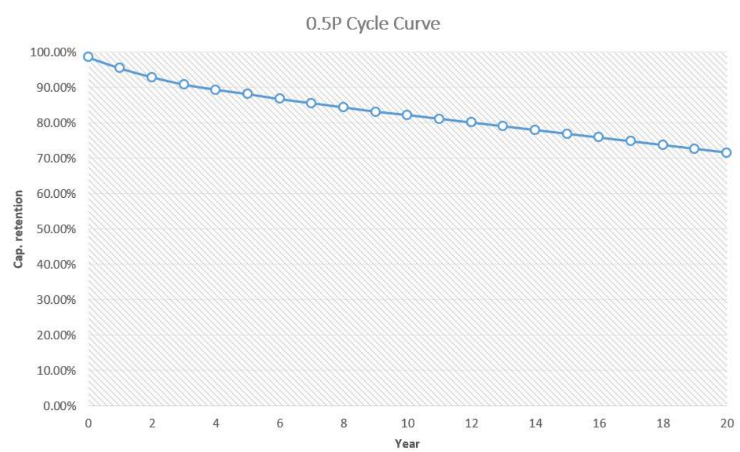

Fig.1 Degradation Curve in 0.25P, 1CPD Working Condition

2.2 Operating Condition: 0.25P, 2 Cycles per Day

Life Cycle Assessment of BESS (2 Cycles/day) - 0.25P

Year	SOH	Cycles
0	98.50%	0
1	93.91%	730
2	90.66%	1460
3	87.97%	2190
4	85.76%	2920
5	83.72%	3650
6	81.78%	4380
7	80.08%	5110
8	78.09%	5840
9	76.12%	6570
10	74.13%	7300

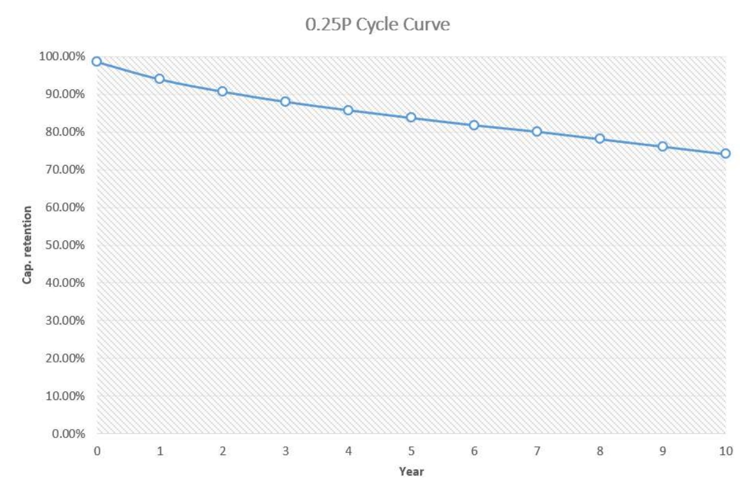

Fig.2 Degradation Curve in 0.25P, 2CPD Working Condition

2.3 Operating Condition: 0.5P, 1 Cycle per Day

Life Cycle Assessment of BESS (1 Cycle/day) - 0.5P

Year	SOH	Cycles
0	98.50%	0
1	95.22%	365
2	92.61%	730
3	90.46%	1095
4	88.91%	1460
5	87.59%	1825
6	86.11%	2190
7	84.88%	2555
8	83.62%	2920
9	82.33%	3285
10	81.32%	3650
11	80.20%	4015
12	79.08%	4380
13	77.96%	4745
14	76.84%	5110
15	75.72%	5475
16	74.60%	5840
17	73.48%	6205
18	72.36%	6570
19	71.25%	6935
20	70.13%	7300

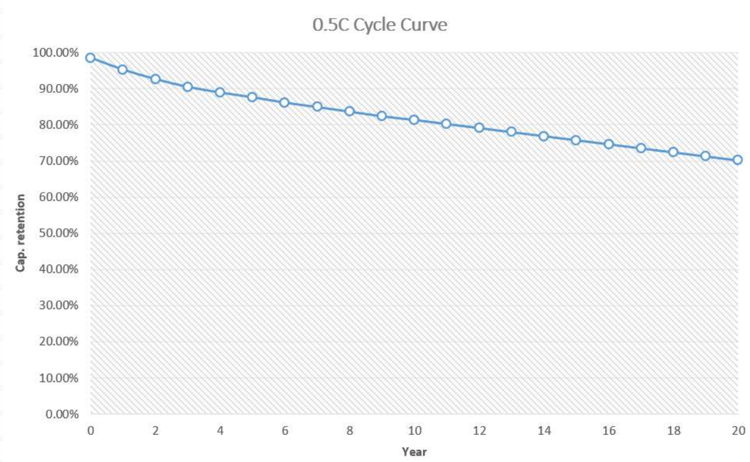

Fig.3 Degradation Curve in 0.5P, 1CPD Working Condition

2.4 Operating Condition: 0.5P, 2 Cycles per Day

Life Cycle Assessment of BESS (2 Cycles/day) - 0.5P

Year	SOH	Cycles
0	98.50%	0
1	93.51%	730
2	90.02%	1460
3	87.12%	2190
4	84.72%	2920
5	82.50%	3650
6	80.41%	4380
7	78.54%	5110
8	76.40%	5840
9	74.27%	6570
10	72.14%	7300

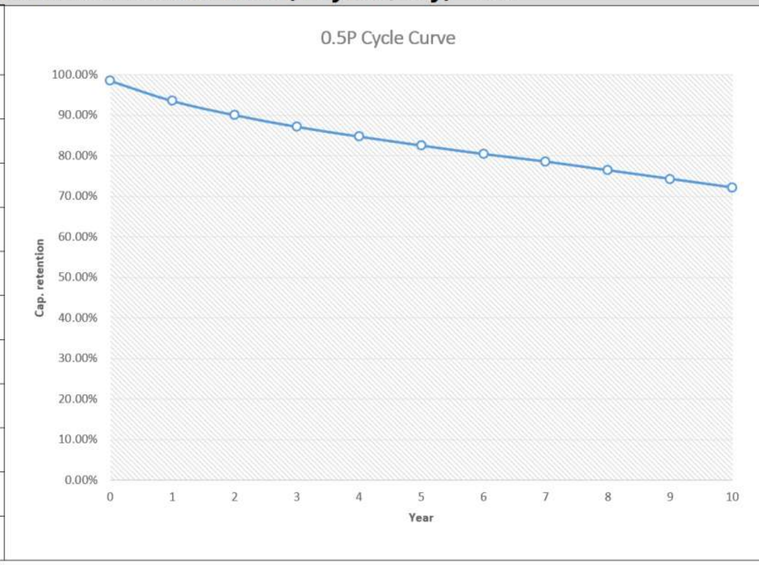

Fig.4 Degradation Curve in 0.5P, 2CPD Working Condition